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Compressing binary strings

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Problem: Can we compress everything, and how much?

erver would know that
nts the first digits of
ntation.
r version of it can be
any short program

```
dByFour = 0
. range(N) :
videdByFour +=
*i+1)
*
```

ree sequences.

11001100110011001100110011001100

Compressibility 1

1/1 point (ungraded)

What is the proportion, among all binary sequences of size 128, that can be compressed down to 100 bits or less?

- ☐ 1/27
- ☒ 1/2²⁷
- ☐ 1/2⁹⁹
- ☐ 1/2¹⁰⁰ − 1/2¹²⁸
- ☐ log₂ (27) − 1
- ☐ 1/100 − 1/128

✓

Answer
Correct:
A sequence of N bits that is compressed exactly by k bits must be represented using $N - k$ bits.
There can be only 2^{N-k} such sequences at most.
We get the maximum number of sequences of N bits that can be compressed at all by m bits *or less* by summing these values up to m .
This makes $2^{N-k+1} - 1$, which means a proportion of 2^{1-k} .
Only a proportion of $1/2^{27} \approx 10^{-8}$ of all binary sequences of size 128 can be compressed down to 100 bits.

Compressibility 2

1/1 point (ungraded)

Consider a given compression method Z . Are there binary sequences of size N that cannot be compressed through Z ? (you may count all compressible sequences.)

☐ Possibly none, for some Z

☒ Yes, at least one, whatever Z

☐ Yes, at least $N/2$



Answer

Correct:

If we count all sequences that are compressible by at least one bit, we get:

$\sum 2^{N-k} = 2^N - 1$. This means that at least one sequence is strictly incompressible for that compressor. The impossibility of compressing all sequences holds even in the case of an "ideal" compressor.

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